Mohamed Aziz Madhbouh

Big Data and Data Analytics Student

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Tunisia

Professional Profile

Third-year Bachelor's student in Big Data and Data Analytics at ISAMM, seeking a final-year internship in generative AI and intelligent application development. Hands-on experience with Python, text processing, classification, and LLM API integration, including Gemini-based build severity prediction. Interested in automated document analysis, structured information extraction, knowledge retrieval, and AI-assisted technical proposal drafting with human review. Experienced in model evaluation, automated testing, and Docker-based development.

Education

Higher Institute of Multimedia Arts of Manouba (ISAMM) *2027*

Bachelor's Degree in Big Data & Data Analytics, 3rd year *Tunisia*

Relevant coursework: distributed systems, Big Data architecture, machine learning, data warehousing, statistical analysis

Projects

CI Log Classifier - Link | *Python, Gemini API, Docker, Jenkins, Grafana, pytest*

- Developed a Python pipeline to process and classify Jenkins build logs into 10 error categories, covering text preprocessing, model training, evaluation, and performance analysis.
- Evaluated the model on 14,320 test logs, achieving 0.77 accuracy and a 0.77 macro F1 score; analyzed out-of-distribution (OOD) data to identify limitations when handling unseen inputs.
- Integrated the Gemini API to predict build severity across four levels (Low, Medium, High, Critical), with automatic switching between models when request limits were reached.
- Integrated the project into a containerized CI monitoring system using Docker, Jenkins, Grafana dashboards, database migrations, and automated tests.

Electric Vehicle Analytics Pipeline - Link | *Hadoop, Spark 3.5, Docker Compose, PySpark*

- Built a distributed data-processing pipeline using HDFS, YARN, and Spark, demonstrating the ability to organize and process structured data at scale.
- Processed a 67 MB vehicle registration dataset with PySpark and generated aggregations by location, manufacturer, model year, vehicle type, and average electric range.

Purchase Analysis with MapReduce - Link | *Python, Hadoop Streaming, MapReduce*

- Implemented mapper and reducer scripts to calculate total sales and transaction counts across 18 product categories from approximately 4.1 million purchase records.
- Validated processing logic locally with Unix pipes before executing jobs on a Hadoop cluster using Hadoop Streaming.

Medical Appointment No-Show Prediction - Link | *Python, Pandas, scikit-learn, Matplotlib, Seaborn*

- Cleaned and explored appointment data, engineered time-based features, and visualized distributions to prepare data for analysis.
- Trained and compared classification models using accuracy, precision, recall, F1 score, and confusion matrices.

Skills

Programming Languages: Python, SQL, Java, R, C/C++ (basic)
Generative AI & Text Processing: LLM API integration (Gemini), text preprocessing, classification, severity prediction, model evaluation
Data Analytics: Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, Jupyter Notebook, feature engineering
Testing, Tools & DevOps: pytest, automated testing, Docker, Docker Compose, Git, GitHub, Jenkins, Grafana, Maven, JDBC
Big Data: Apache Hadoop, Apache Spark, PySpark, HDFS, YARN, MapReduce